

CSCE 753 COMPUTER VISION · MODULE 06 · LECTURE NOTES

Stereo and Dense Depth

A depth value at every pixel.

MODULE OUTCOMES

- Explain rectification and why it is done first.
- Explain the cost volume and the matching cost choices.
- Explain why smoothness is required and how it is imposed.
- Compare passive stereo against active depth sensing.
- State where dense stereo fails and why.

1 Rectification and the cost volume

RECTIFY FIRST, AND THE SEARCH BECOMES A SCAN ALONG A ROW

Module 04 reduced the correspondence search to a line. Rectification warps both images with homographies so that line is *the same horizontal row* in both images.

Then the match for pixel (u, v) in the left image lies at $(u - d, v)$ in the right, for some disparity $d \geq 0$ — a one-dimensional search over integers, independently per pixel.

Which makes the computation cache-coherent and vectorisable, and that is the actual reason it is done: the geometry does not require rectification, and the arithmetic is far faster with it. **CSCE 735's memory-locality lesson, in a vision pipeline.**

Depth follows from disparity directly: $Z = fb/d$. Disparity and inverse depth are therefore the same quantity up to a constant factor, which is why stereo methods work in disparity throughout, and why uniform disparity sampling is a sensible choice while uniform depth sampling is not — it matches where the precision actually lies (Module 04 §4's Z^2 relationship).

For each pixel (u,v) and each candidate disparity d , compute a MATCHING COST:

$$C[v][u][d] = \text{cost of matching left}(u,v) \text{ against right}(u-d, v)$$

That three-dimensional array is the COST VOLUME, and it is the common structure of classical and learned stereo alike.

MATCHING COSTS, weakest to strongest:

SAD / SSD	sum of absolute or squared differences. Fast. Breaks under any exposure difference between the two cameras, which real rigs always have.
NCC	normalised cross-correlation. Invariant to affine intensity change (gain and offset).
CENSUS	compare each neighbour's intensity to the window centre, producing a bit string; cost is the Hamming distance between the two strings. INVARIANT TO ANY MONOTONIC intensity transform, and very cheap. The best classical choice, and underused.
LEARNED	a small CNN scores patch pairs. Better still, especially on weak texture.

WINNER-TAKE-ALL over d gives a disparity map that is recognisable and very noisy. Everything after this point is about the noise.

2 Smoothness

PER-PIXEL MATCHING IS AMBIGUOUS, SO A PRIOR IS REQUIRED

A textureless region matches equally well at every disparity. The cost volume is flat along d and the argmin is determined by sensor noise — which is Module 01's under-determination appearing per-pixel.

A repeated pattern matches well at several disparities, and winner-take-all selects one of them arbitrarily and confidently. **The failure is worse than the textureless case**, because the cost at the chosen disparity is genuinely low, so confidence measures based on cost value do not catch it.

So impose smoothness: neighbouring pixels probably have similar disparity, because surfaces are mostly continuous. This is a prior, chosen because the inversion needs one.

But it is false at object boundaries, where disparity jumps discontinuously. **So the smoothness penalty must be edge-aware** — weak across image intensity edges, strong within uniform regions. That single refinement is most of what separates good stereo from bad, and it is the same insight as the edge-aware filtering of CSCE 748 Module 03.

Method	How it works	Note
Window aggregation	Sum the matching cost over a patch rather than a pixel.	Smoothness imposed implicitly, and it blurs depth boundaries — a larger window means less noise and worse edges.
Graph cuts	Minimise a global energy of data plus smoothness terms, solved exactly for two labels and approximately for more.	Minimum cut — CSCE 669 Module 11 §1's reduction, used in anger. Accurate and slow.
Semi-global matching (SGM)	Dynamic programming along 8 directions through the image, with the costs summed.	The classical sweet spot. Approximates the global solution, runs in real time, around a hundred lines. Implement this one first.
Bilateral / guided aggregation	Aggregate the cost volume with an edge-aware filter.	Cheap, parallel, and surprisingly effective; the filter does the edge-awareness for you.
Learned	3D convolutions over the cost volume, trained end to end.	Best accuracy on benchmarks , requires training data with ground-truth depth, and inherits Module 12's distribution-shift problem.

3 Active depth sensing

IF THE SCENE HAS NO TEXTURE, SUPPLY SOME

Structured light projects a known pattern — stripes, dots, or a pseudorandom speckle — and triangulates from how the pattern deforms. A blank wall becomes a textured wall, and the correspondence problem becomes easy because the projected pattern is designed to be locally unique.

Time of flight measures the round-trip delay of modulated light per pixel, so there is no correspondence problem at all — depth is measured directly rather than inferred.

Each has characteristic failures. Structured light is overwhelmed by sunlight (the projector cannot compete) and fails on dark surfaces (insufficient return) and at range (pattern divergence). Time of flight suffers multi-path error at concave corners, where light arrives by two routes, and interference between multiple units in the same room.

And both fail on the same materials as passive stereo. Glass, polished metal, and water defeat every method in this module, because they all assume a surface returns light from the place where the surface is — and a mirror returns light from somewhere else entirely, which is a correct measurement of the wrong geometry. **No amount of sensing resolves it; it needs a prior about materials** (Module 01 §3's fourth strategy).

4 Where dense stereo fails

- **Textureless surfaces.** No signal to match against. **Painted walls, skies, tabletops, doors. The dominant failure indoors,** and the reason consumer depth cameras are active rather than passive.
- **Repeated texture.** Confident wrong matches, as §2 described. **Brick, tile, railings, keyboards, textiles.**
- **Occlusion.** Pixels visible in one view and not the other **have no correct disparity at all** — there is nothing to match. **They must be detected and labelled invalid, not estimated,** and a left-right consistency check is the standard detector.
- **Specular and transparent surfaces.** Apparent brightness depends on viewpoint, so the brightness-constancy assumption underlying every matching cost fails outright rather than degrading.
- **And distance. Error grows as Z^2** (Module 04 §4), so every rig has a usable range that is computable before it is built, and operating beyond it produces depth maps that look plausible and are meaningless.
- **So report a confidence or validity mask with every depth map. A depth value in a textureless region is fabrication** — the algorithm had no information and returned a number anyway — and downstream code must be able to distinguish 'measured 2.3 m' from 'guessed 2.3 m'. **Systems that propagate depth without validity fail in ways that are extremely hard to trace,** because the error enters as a plausible measurement.

MULTI-VIEW STEREO IS THE SAME IDEA WITH MORE VIEWS

With n calibrated views, build the cost volume by projecting each candidate depth into every view and measuring photo-consistency across all of them. The structure is identical; only the cost's support changes.

More views resolve the ambiguities that two cannot. A repeated pattern is unlikely to be consistent across six genuinely different viewpoints, and a textureless region at least becomes better constrained by its boundaries. **This is Module 01's first strategy applied harder**, and it is the only one of the four with guarantees.

And occlusion becomes tractable rather than fatal, because a point hidden in some views is visible in others — provided the view selection accounts for visibility, which is the main engineering difficulty.

This is what COLMAP's dense reconstruction stage does, and it is the classical method that **Module 11's neural representations displaced** — by optimising a differentiable renderer against the photographs instead of searching a cost volume. **Same problem, same input, entirely different formulation**, and the comparison is the most instructive thing in this course.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 12 (free PDF)	Stereo correspondence, rectification, and the cost volume — the reference for §1 and §2.
Hirschmüller — Stereo Processing by Semi-Global Matching (free)	SGM, the method of §2's table . Implementable directly from the paper.
Scharstein & Szeliski — the Middlebury stereo benchmark (free)	Ground-truth data and a leaderboard , plus the taxonomy that organises §2's methods.
Zabih & Woodfill — Non-parametric Local Transforms (census) (free)	The census transform of §1 , and the monotonic-invariance argument.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Rectify a calibrated stereo pair** and verify that matched features land on the same row.
- 2 **Build a cost volume with SAD, NCC, and census** and compare the winner-take-all disparity maps.
- 3 **Change the exposure on one camera** and report which cost survives.

- 4 **Implement SGM** and compare against winner-take-all on the same cost volume.
- 5 **Make the smoothness penalty edge-aware** and measure the improvement at depth boundaries.
- 6 **Implement a left-right consistency check** and visualise the occlusion mask.
- 7 **Measure the fraction of pixels your method should refuse** on a real indoor scene.
- 8 **Compare your depth against an active depth camera** if you have one, and locate where each fails.
- 9 **Photograph a glass or chrome object** with both and report what each returns.
- 10 **Verify the Z^2 error law** on objects at measured distances.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why rectify, given the geometry does not require it?
- 2 Why are disparity and inverse depth the same quantity?
- 3 What is the cost volume, and which matching cost is the best cheap classical one and why?
- 4 Why is per-pixel matching insufficient, and why must smoothness be edge-aware?
- 5 Compare five ways of imposing smoothness.
- 6 What do structured light and time of flight each do, and how does each fail?
- 7 Which materials defeat every method here, and why?
- 8 Name five stereo failure modes.
- 9 Why must a depth map carry a validity mask?